Generic1 model

Parametrization

The Type 1 generic model implements the following precision matrix

$$\mathbf{Q} = \tau(\mathbf{I} - \frac{\beta}{\lambda_{max}}\mathbf{C})$$

where $\mathbf{C}$ is the structure matrix. The parameter λ_{max} is the maximum eigenvalue of $\mathbf{C}$, which allows β to be in the range $\beta \in [0, 1)$

Note that λ_{max} is computed unless given, which can be very time consuming for a large dimension.

Hyperparameters

The two parameters of the generic1 model are represented as

$$\begin{aligned} \theta_1 &= \log(\tau) \\ \theta_2 &= \text{logit}(\beta) \end{aligned}$$

and priors are assigned to (θ_1, θ_2)

Specification

The generic1 model is specified inside the `f()` function as

```
f(<whatever>,model="generic1",Cmatrix = <Cmat>, hyper = <hyper>)
```

`<Cmat>` can be given in two different ways:

- a dense matrix or a sparse-matrix defined be `Matrix::sparseMatrix()`.
- the name of a file giving the structure matrix. The file should have the following format

  $$i \quad j \quad \mathbf{C}_{ij}$$

  where i and j are the row and column index and $\mathbf{C}_{ij}$ is the corresponding element of the precision matrix. Only the non-zero elements of the precision matrix need to be stored in the file.

Hyperparameter spesification and default values

doc `A generic model (type 1)`

hyper

theta1

hyperid `19001`
name `log precision`
short.name `prec`
prior `loggamma`
param `1 5e-05`
initial `4`
fixed `FALSE`

to.theta `function(x) log(x)`

from.theta `function(x) exp(x)`

theta2

hyperid `19002`

name `beta`

short.name `beta`

initial `2`

fixed `FALSE`

prior `gaussian`

param `0 0.1`

to.theta `function(x) log(x / (1 - x))`

from.theta `function(x) exp(x) / (1 + exp(x))`

constr `FALSE`

nrow.ncol `FALSE`

augmented `FALSE`

aug.factor `1`

aug.constr

n.div.by

n.required `TRUE`

set.default.values `TRUE`

pdf `generic1`

Example

```
n = 100
## build a structure matrix
Cm = matrix(runif(n^2,min=-1,max=1),n,n)
diag(Cm) = 0
Cm = 0.5*(Cm + t(Cm))
lambda.max = max(eigen(Cm)$values)

## define the precision matrix
beta = 0.9
Q = diag(rep(1,n)) - beta/lambda.max * Cm
Sigma = solve(Q)

#simulate data
require(mvtnorm)
sd = 0.001
z = rnorm(n)
eta = rmvnorm(n=1,sigma = Sigma)
y = c(eta) + sd*rnorm(n) + z
idx = 1:n
d = list(y=y,idx=idx,z=z)
```

```
## Alternative 1
## print the file containing the C matrix
file = "Cmatrix.dat"
cat("",file=file, append = FALSE)
for(i in 1:n)
{
    j = i
    cat(i, j, Cm[i,j], "\n", sep = " ", file=file, append=TRUE)
    if (i < n)
        for(j in (i+1):n)
            cat(i, j, Cm[i,j], "\n", sep = " ", file=file, append=TRUE)
}
formula = y ~ f(idx, model = "generic1", Cmatrix = file) + z

## Alternative 2
## formula = y ~ f(idx, model = "generic1", Cmatrix = Cm) + z

## Alternative 3
## formula = y ~ f(idx, model = "generic1", Cmatrix = as(Cm, "dgTMatrix"))+z

######################################################################

result = inla(formula, data=d,family="gaussian",
              control.family = list(initial = log(1/sd^2), fixed=TRUE),
              verbose=T, keep=T)
```

Notes

None